LINEAR PATH

5

- Has un-normalised density:

$$\begin{aligned}\gamma_\beta(x) &= \eta(x)^{1-\beta}\gamma(x)^\beta \\ &= \eta(x)w(x)^\beta, \qquad w(x) = \frac{\gamma(x)}{\eta(x)}\end{aligned}$$

- Equivalently, it corresponds to the linear interpolation between the log-densities

$$
\begin{aligned}
V_\beta &= (1-\beta)V_0 + \beta V_1 \\
&= V_0 + \beta V, \qquad V = \log w
\end{aligned}
$$

- The canonical choice is the linear/geometric path

$$\pi_\beta \propto \eta^{1-\beta}\pi^\beta$$

- The linear path flattens the likelihood ratio between referece and target

$$\frac{d\pi_\beta}{d\eta}(x) \propto \left(\frac{d\pi}{d\eta}(x)\right)^\beta \propto w(x)^\beta$$

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